Test Sheet

2026 Raleigh Division B Invitational, Presented by NCBioNetwork

Remote Sensing

DO NOT WRITE ON THIS TEST

Open this booklet when the event leader starts time

Tiebreakers: 6

1. What is remote sensing? (1 pt)

2. Match each term with its correct definition (5 pts):

   A. Radar
   B. Scatterometer
   C. Sounder
   D. Lidar
   E. Laser altimeter

Definitions:

1. A sensor that measures atmospheric conditions vertically by detecting or emitting signals to analyze properties such as temperature or humidity at different heights.

2. A system that determines distance using light impulses and calculates target location by multiplying travel time by the speed of light.

3. A device that measures elevation using lidar technology.

4. A sensor that uses radio signals to measure distance by emitting impulses and detecting the energy that scatters back from an obstacle.

5. A device designed to measure the amount of backscattered radiation.

3. What does ALI stand for? (2 pts)

   A. Airborne Light Imager

   B. Advanced Land Imager

   C. Atmospheric Luminance Instrument

   D. Active Laser Interferometer

4. Which of the following types of data is NOT provided by NASA's TERRA satellite? (2 pts)

A. Land surface temperature and vegetation cover

B. Atmospheric pollution and cloud properties

C. Ocean color and sea surface temperature

D. Interior structure of the Earth's crust

5. Give the ranges in **kilometers** for each of the following orbits (3 pts):

- Low-Earth: ___a____ to ____b___ above Earth
- Medium-Earth: _____c____ to ______d____ above Earth
- High-Earth: ______e_____ above Earth

6. Which of the following correctly explains why Sun-synchronous polar-orbiting platforms like NASA's Aqua are valuable for climate monitoring? (2 pt)

A. They remain fixed above one location, allowing continuous observation of regional weather systems.

B. They orbit at the same altitude as geostationary platforms, enabling high temporal resolution.

C. They pass over the same location at the same solar time each orbit, ensuring consistent lighting conditions for global, repeatable observations.

D. They orbit only over the equator, minimizing atmospheric distortion in data collection.

7. During a major cyclone, thick clouds cover a river basin for several days. You must map the flood extent over a large area.

- Would you choose a passive optical sensor (visible/NIR/SWIR) or an active microwave (SAR) sensor? (1 pt)
- Name the relevant part(s) of the electromagnetic spectrum and give two reasons for your choice grounded in the physics of EM waves–surface/atmosphere interaction. (4 pts)

8. Which of these is an active EM sensor? (1 pt)

a. Spectrometer
b. Accelerometer
c. Hyperspectral Radiometer
d. Radar

9. A satellite image shows surface reflectance of a crop field: Red = 0.18, NIR = 0.62.

(a) Compute NDVI. (1 pt)

(b) Briefly interpret the value in terms of vegetation condition. (2 pt)

10. Which is not a satellite? (1 pt)

a. Aqua
b. SeaWiFS
c. Cloudsat
d. Terra

11. Which type of orbit is best for real-time weather monitoring, and why? (2 pts)

12. Which instrument type (optical scanner, multispectral imager, radiometer, or radar altimeter) is best for measuring sea surface height, and why can't an optical sensor do this? (2 pts)

13. Give one advantage of a pushbroom scanner over a whiskbroom scanner in satellite imaging. (2 pts)

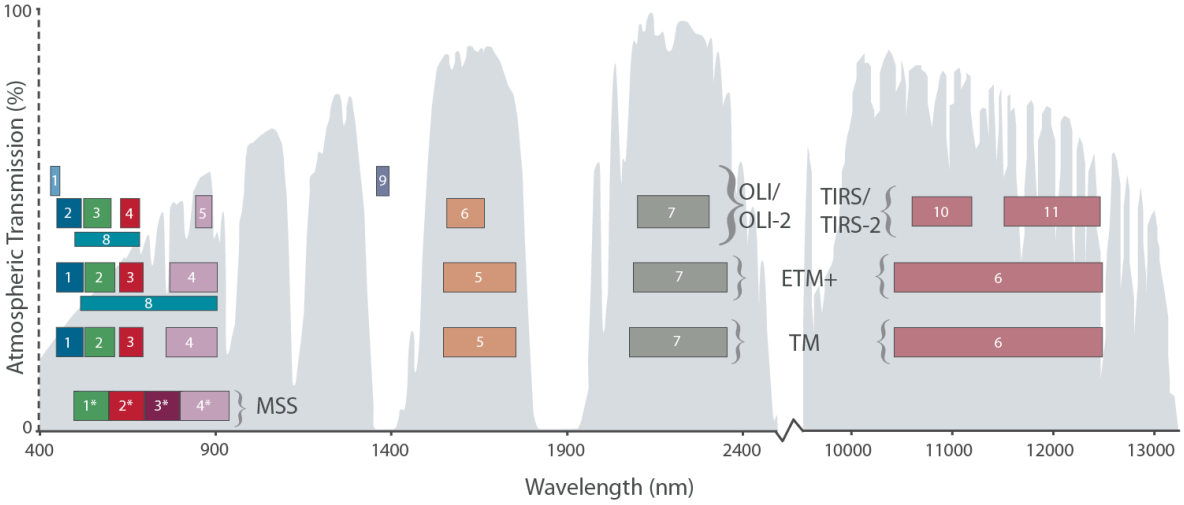
14. What is the compass bearing for an Azimuth of 212°? Answer in quadrant bearing. (1 pt)

15. (4 pts)



What causes the large green crop circles seen in satellite images taken by Landsat TM of southern Alberta? (Tiebreaker Question #1)

Use the graph below to answer questions 16-20:



16. Why are there no Landsat bands in the 1400 and 1900 nm areas of the spectrum? (2 pts)

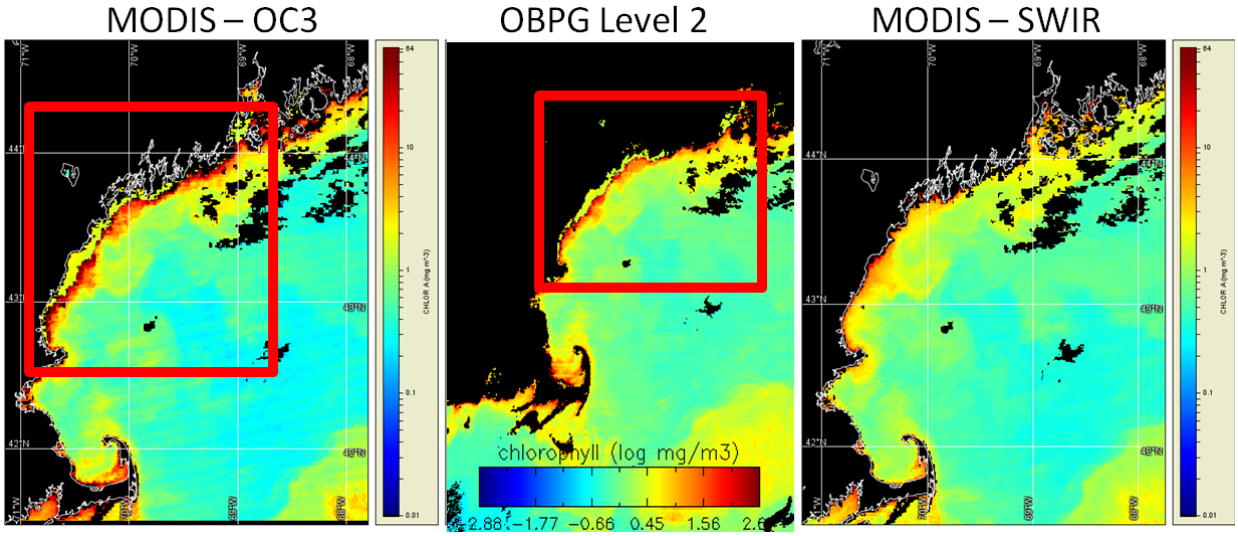
17. Which Landsat sensor was the first to include a widely used thermal infrared band, and at approximately what wavelength does it operate? (4 pts)

18. What is one improvement that Landsat OLI has compared to the older MSS sensor? (1 pt)

19. Which Landsat band(s) would be most appropriate for distinguishing snow from clouds, and why? (3 pt)

20. Why are the thermal bands shown far to the right of the chart, past 10,000 nm? (2 pts)

21.



The three MODIS chlorophyll maps below were produced using different processing methods. In the boxed coastal region, the left image (OC3) shows much higher chlorophyll values than the right image (SWIR). Which of the following best explains why the OC3 method overestimates chlorophyll in this area? (2 pts) (Tiebreaker Question #2)

A. OC3 cannot detect chlorophyll in shallow water.

B. The OC3 algorithm uses visible blue-green bands, which are strongly affected by sediment and dissolved organic matter near the coast.

C. The SWIR method removes chlorophyll signals, so it shows lower values everywhere.

D. The OBPG Level-2 product is the only one that measures real chlorophyll concentration; the others are false-color images.

Use the information below to answer questions 22-27 (14 pts):



The image above shows annual actual evapotranspiration (ETa) in the Modesto, California area for the year 2020. The map was created using data from NASA and the U.S. Geological Survey's Landsat satellites, analyzed through the SSEBop (Simplified Surface Energy Balance operational) model by the Earth Resources Observation and Science (EROS) Center.

Evapotranspiration (ETa) measures the total amount of water that leaves the land surface and plants through evaporation (from soil and water) and transpiration (from plants). ETa helps scientists and farmers understand how much water crops actually use.

In the image, colors represent total annual ETa: Tan/yellow = low ETa (little or no vegetation, urban areas), Green to dark blue = higher ETa (actively growing vegetation or irrigated farmland).

The city of Modesto appears near the center. Surrounding areas show agricultural fields, many of which are vineyards and orchards. Researchers, including those at Gallo Winery, use Landsat data to optimize irrigation, reduce water use, and maintain crop quality—helping conserve groundwater in California's Central Valley.

22. Which areas in the Modesto image likely correspond to farmland, and which to urban land? Explain using color and evapotranspiration clues. (4 pts)

23. Landsat uses a thermal infrared sensor to estimate ETa. Describe what warmer vs. cooler surface temperatures indicate about water use. (2 pts)

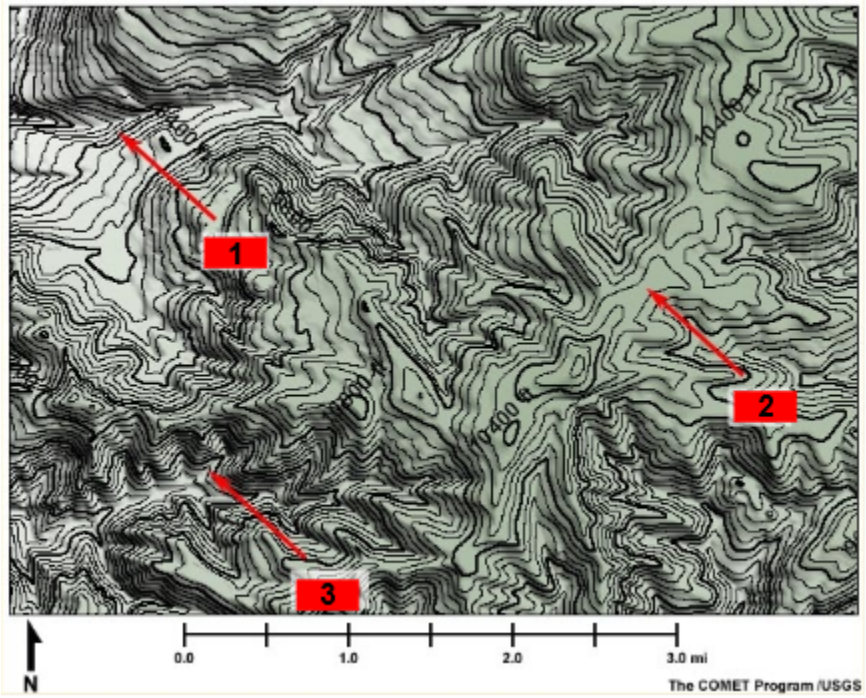
24. Explain why vegetated farmland shows higher ETa than bare or urban land, in terms of the surface energy balance. (2 pts)

25. Landsat's pixel size is 30 meters. Why is this spatial resolution useful for monitoring individual agricultural fields? (1 pt)

26. How can satellite-based ETa maps help farmers in drought-prone regions manage irrigation? (3 pts)

27. Landsat Next will collect thermal data more frequently and at higher resolution. Explain how this improves environmental monitoring. (2 pts)

28. Identify the indicated area (1 pt each)



1. _____

2. _____

3. _____

29. When there are large amounts of smoke or dust (aerosols) in the atmosphere, how does this most likely affect satellite images taken in the visible spectrum? (1 pt)

A. The images become brighter because more sunlight reaches the surface.

B. The images appear blurrier or hazier because particles scatter sunlight.

C. The surface looks warmer because aerosols trap heat.

D. The resolution of the satellite sensor decreases permanently.

30. Two fields appear side-by-side in a Landsat thermal image. Field A has a high albedo (light color) and Field B has a low albedo (dark color). Which statement correctly describes their surface temperatures? (2 pts)

A. Field A will likely be cooler than Field B because it reflects more sunlight.

B. Field A will likely be warmer than Field B because it absorbs more energy.

C. Both fields will have the same surface temperature because albedo doesn't affect heat.

D. The darker field will scatter more light, reducing its temperature.

31. A research team compares two Landsat 8 time-series datasets of farmland from June and September.

They analyze three bands:

- Red (0.64–0.67 μm)
- Near-Infrared (NIR, 0.85–0.88 μm)
- Shortwave Infrared (SWIR, 1.57–1.65 μm)

They find that from June to September:

- NIR reflectance decreases
- Red reflectance increases slightly
- SWIR reflectance increases significantly

Which of the following best explains the environmental change that occurred in these fields? (3 pts) (Tiebreaker Question #3)

A. The plants grew more vigorously and absorbed less sunlight, indicating a healthy crop.

B. The soil became wetter after irrigation, reducing reflectance in all infrared bands.

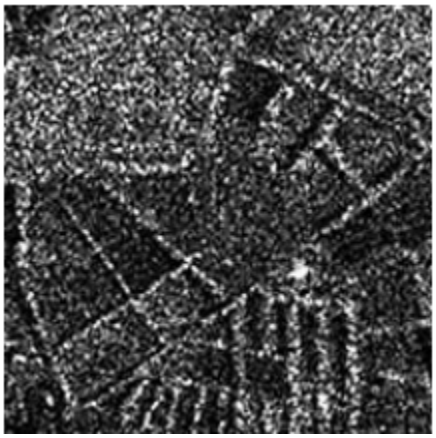
C. Vegetation dried and thinned, causing more reflected SWIR and Red and less NIR as water content decreased.

D. Atmospheric haze increased, artificially lowering the NIR signal while leaving SWIR unchanged.

32. What does MODIS stand for? (1 pt)

33. For MODIS, which bands are used to make a true-color image? (3 pts)

34. Fill in the blank: Passive remote sensors measure radiant energy _____ by the Earth's atmosphere system. (1 pt)

35. 

The radar image above shows a grainy "salt-and-pepper" texture known as speckle. Speckle occurs in Synthetic Aperture Radar (SAR) images because each radar pixel records the combined return from many small surface scatterers (rocks, buildings, vegetation).

Which statement best explains the nature of radar speckle and how scientists reduce its effect? (2 pts)

A. Speckle is caused by random noise in the radar instrument; it can be removed completely by increasing radar power.

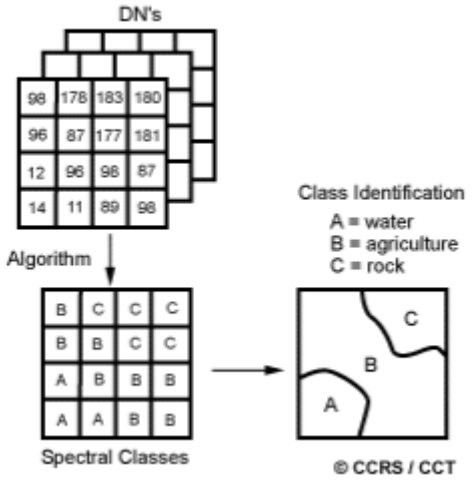
B. Speckle results from constructive and destructive interference of radar waves from many scatterers within one pixel; it can be reduced by averaging multiple looks or using filters.

C. Speckle appears when the radar wavelength is shorter than visible light; it disappears if the satellite switches to optical imaging.

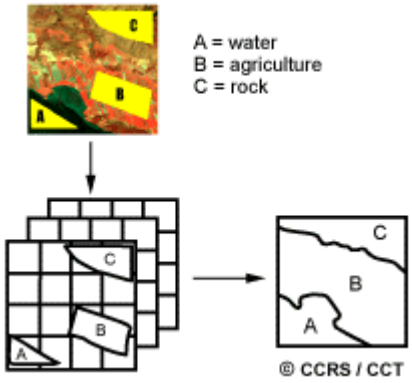
D. Speckle indicates areas of strong backscatter due to metallic objects; bright speckles always mark man-made surfaces.

36. What does the term “albedo” refer to? (1 pt)

37. A.



B.

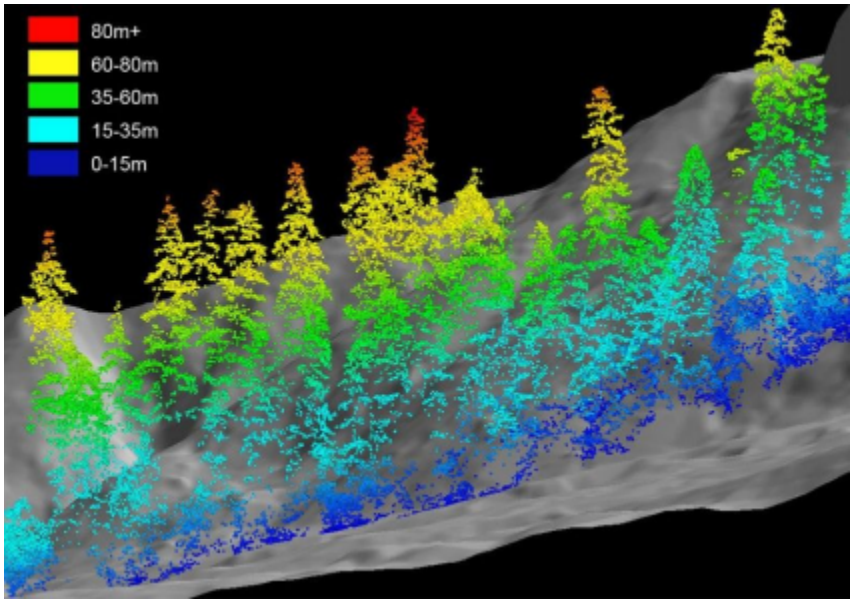


What are processes A and B formally known as? (1 pt each)

38. What are the two most commonly used geographic systems? (1 pt)

39. What was the Terra satellite originally known as? (1 pt)

Use the image to answer the following questions:



40. What instrument was used to make this image? (2 pts)

41. What do each of the dots in the image represent? (1 pt)

42. What is the most significant natural source of aerosols? (1 pt)

a) The ocean

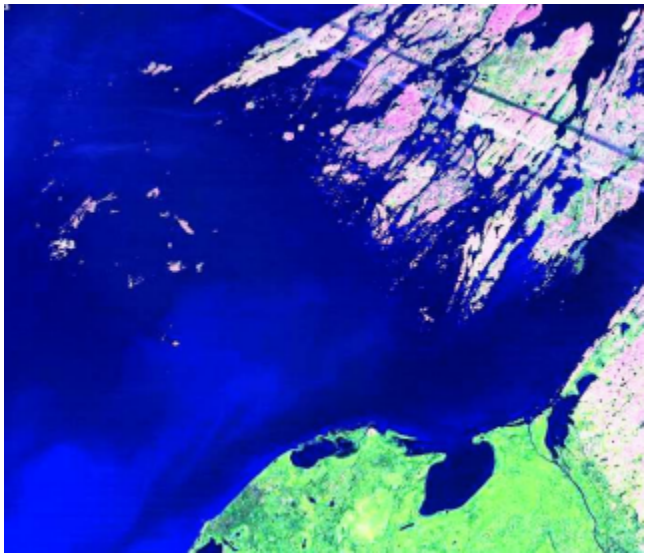
b) Volcanoes

c) Wind

d) Wildfire

e) None of the above

Use the image below to answer the following questions 43-45 (2 pts each):



43. What is the diagonal white line at the top right of the image?

44. Was Landsat 5 a low, medium, or high Earth orbit satellite?

45. What sensors did Landsat 5 have?

46. List and define the four spectral classes of optical remote sensing systems. (8 pts) (Tiebreaker #4)

47. When the atmosphere contains a high concentration of aerosols (for example, after wildfires), how would this most likely affect a visible-light satellite image of Earth's surface? (3 pts)

A. The image becomes brighter and clearer because aerosols reflect more sunlight.

B. The image appears hazier and less detailed because scattered light reduces surface contrast.

C. The image shows warmer temperatures due to increased surface heating.

D. The image becomes sharper since aerosols filter out unwanted wavelengths.

48. What is the name of the scientist who first proposed the astronomical theory for climate change? (1 pt)

49. The marginal information of a topographic or satellite map usually includes which of the following? (2 pts)

A. Only the latitude and longitude of the map center

B. Aerial photograph of the area at high resolution

C. Contour lines and symbols for elevation

D. Metadata such as scale, legend, title, date, and projection information

50. A Landsat image has a scale of 1:100,000. If two towns are 5 cm apart in the image, what is their actual distance on the ground? (1 pt)

51. Which satellite sensor and band combination would best detect active lava flows during a volcanic eruption? (2 pts)

A. Landsat TM Band 3 (red, 0.63–0.69 μm)

B. MODIS thermal IR (3.9 μm and 11 μm)

C. Sentinel-2 green band (0.56 μm)

D. Landsat OLI coastal/aerosol band (0.43 μm)

52. Which of the following satellites or instruments listed below would be most useful for creating a global digital elevation model (DEM)? (2 pts)

A. Terra (MODIS instrument)

B. Sentinel-2 (multispectral optical imaging)

C. Shuttle Radar Topography Mission (SRTM)

D. Landsat 9 (OLI sensor)

53. What does the term “climate noise” refer to? (2 pts)

54. Which Landsat TM band is most useful for assessing vegetation health and biomass by detecting near-infrared reflectance from plant leaves? (3 pts)

A. TM 2 (green)

B. TM 3 (red)

C. TM 4 (near infrared)

D. TM 7 (shortwave infrared)

55. If a scientist wants to map coastal regions and shallow water depth (bathymetry), which TM band would provide the most useful data? (3 pts)

A. TM 1 (blue)

B. TM 3 (red)

C. TM 5 (shortwave IR)

D. TM 6 (thermal IR)

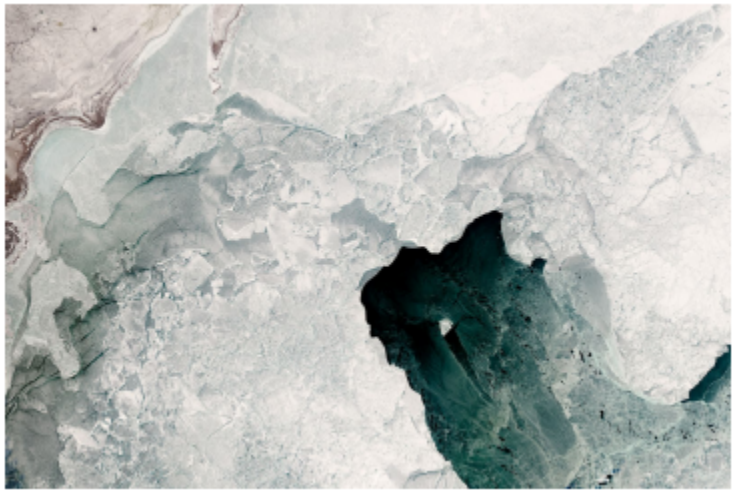
56. A Landsat image shows areas of low reflectance in the near IR (TM 4) and high brightness in the thermal IR (TM 6). What does this combination most likely indicate? (3 pts)

A. Healthy, dense vegetation

B. Vegetation stress or drought conditions

C. Cloud cover interference

The following image of the northern Caspian sea was taken by OLI. Use this to answer the questions below:



57. What does OLI stand for? (1 pt)

58. What type of scanner is OLI-1? (1 pt)

59. What satellite does OLI-1 fly on? (2 pts)

60. The northern areas of the Caspian Sea are more prone to freezing in wintertime. Why? (3 pts) (Tiebreaker #5)

61. The National Land Cover Database (NLCD) provides standardized land cover maps for the contiguous United States (CONUS). What type of information does it primarily show? (3 pts)

A. Elevation and terrain slope

B. Land cover categories such as forest, urban, water, grassland, and agriculture

C. Air temperature and humidity patterns

D. Ocean salinity and wave height

62. When radar waves strike a surface, the return signal can be measured in different polarizations (HH, HV, VH, VV). Why might a cross-polarized (HV or VH) return be stronger over a forest canopy than over smooth open water? (2 pts)

63. A satellite sensor with an 8-bit radiometric resolution can record 256 levels of brightness. If a newer sensor increases this to 12 bits, how many distinct intensity levels can it detect? (2 pts)

64. A topographic map uses a UTM coordinate grid, with easting = 500 000 m E and northing = 4 000 000 m N. What projection system is UTM based on, and why is it preferred for remote-sensing imagery over large areas? (2 pts) (Tiebreaker #6)